

No. of Printed Pages : 4
Roll No.

221763

6th Sem/ Mech. Mech (Tool & Die Design)
Subject : Metrology & Quality Control

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 The instrument used to measure angle relative to horizontal plane is. (CO2)

- a) Clinometer b) Comparator
- c) Dial Gauge d) Ring Gauge

Q.2 External taper can be measured accurately by (CO2)

- a) Micrometer
- b) Vernier Caliper
- c) Sine Bar and Slip Gauge
- d) Plug Gauge

Q.3 Straightness may be checked by (CO3)

- a) Straight Edge b) Spirit Level
- c) Auto Collimator d) All of the above

- Q.4 Which of the following is a control chart for variables. (CO4)
- a) X-Chart b) C-Chart
 - c) U-Chart d) All of the above
- Q.5 What is the full form of TQM? (CO5)
- a) Total Quality Management
 - b) Total Quality Method
 - c) Team Quality Management
 - d) Team Quality Measurement
- Q.6 Thermocouple are ____ transducer. (CO6)
- a) Active b) Passive
 - c) Adhesive d) None of the above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define Measurement. (CO1)
- Q.8 Define Kaizen. (CO2)
- Q.9 What is the use of Gauges? (CO2)
- Q.10 Define Error. (CO3)
- Q.11 Write the formula for Mean Deviation. (CO4)
- Q.12 Define Transducer. (CO6)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Write the objectives of inspection? (CO1)
- Q.14 Explain Plug Gauge and Ring Gauge with neat sketch. (CO2)
- Q.15 Write the advantages of Mechanical Comparator. (CO3)
- Q.16 Explain Sine Bar with diagram. (CO4)
- Q.17 Write the Characteristics of Single Sampling Plan. (CO5)
- Q.18 Explain the principles of TQM. Write the benefits of TQM. (CO5)
- Q.19 Write short note of Preventive and Operative inspection. (CO1)
- Q.20 Explain Resistive Potentiometer. (CO5)
- Q.21 Write short note on (CO3)
- a) Straight Edge b) Spirit Level
- Q.22 Explain Cause and Effect diagram with neat sketch. (CO4)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain four types of QC tools. (CO5)

Q.24 Explain LVDT with neat sketch. (CO6)

Q.25 Explain the following control charts. (CO4)

a) X-Chart

b) R-Chart

c) P-Chart

d) C-Chart